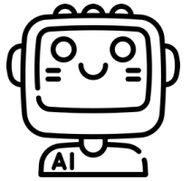
Problem: Find minimum of $\frac{a}{b^3+4}+\frac{b}{c^3+4}+\frac{c}{d^3+4}+\frac{d}{a^3+4}$ given $\boldsymbol{a, b, c, d}$ are non-negative and $\boldsymbol{a+b+c+d=4}$.



Base-LLM

✓ Reasoning Steps

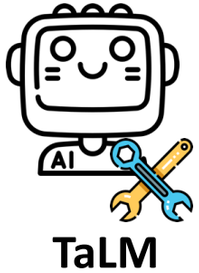
✓ Final solution

Base-LLM generation

Step 1. An interior critical point. By symmetry and by the method of Lagrange multipliers one finds the only symmetric critical point is $a = b = c = d = 1$, at which…

Step 2. Checking the boundary. On boundary at least one variable is zero… hence **smallest boundary value is $\frac{2}{3}$**

… By cyclic symmetry, this minimum is when exactly two consecutive variables are 0 and the other two are 2.



TaLM

✕ Tool-induced Myopia

✓ Final solution

TaLM generation

Here are a few **observations**:

Observation 1. If exactly one of the variables is nonzero, then: $S = 4\times\left(\frac{1}{0^3+4}\right) = 1.$

Observation 2. If all four variables are equal, then: $S = 4\times\left(\frac{1}{1^3+4}\right) = 0.8.$

Observation 3. **[Tool Call 1] A straightforward numerical check** shows minimum occurs at $a = 2$, which gives $\frac{2}{3}$.

Observation 4. **[Tool Call 2] Checking other patterns** (3 nonzeros, random sampling) never yields a smaller value than $\frac{2}{3}$. Thus**, the global minimum is $\frac{2}{3}$**.